AI-Agent Foundry

The **AI-Agent Foundry** is StatusNeo's enterprise AI portfolio designed to help organizations accelerate digital transformation through autonomous systems, intelligent automation, and domain-specific AI solutions.

It combines:

- Production-ready AI platforms
- Industry-specific vertical solutions
- Autonomous super agents
- SDLC automation agents

The Foundry enables enterprises to move beyond experimentation and deploy **secure, scalable, explainable AI systems in production environments**.

Portfolio Structure

The AI-Agent Foundry is organized into four strategic layers:

Layer	Purpose	Offerings
Horizontal Accelerators	Cross-industry SaaS AI platforms	Testkraft, NeoLens, S^2 Studio Synthesis, Genesis
Vertical Solutions	Industry-focused AI systems	StatusNeo Voice Agent, AML Monitoring Agent
Super Agents	High-capability autonomous agents	Debugging Expert Agent
SDLC Agents	Development lifecycle automation	Code Review Agent

Horizontal Accelerators

1. Studio Synthesis

Enterprise AI Platform for Automated Design-to-Code Transformation

Executive Summary

Studio Synthesis is an enterprise-grade AI platform that automates the transformation of Figma design assets into production-ready, full-stack applications. It enables organizations to reduce

development cycle time, improve engineering consistency, and embed testing and governance into the earliest stages of software creation.

The platform addresses a structural inefficiency in modern product development: the manual translation of design specifications into functional code. By programmatically generating frontend architecture, backend APIs, database schemas, and automated test suites, Studio Synthesis enables engineering teams to move from approved design to deployable application in significantly less time.

Built for enterprise environments, Studio Synthesis integrates natively with GitHub and connects seamlessly to Slack, Jira, and Microsoft Teams, ensuring alignment with existing DevOps, Agile, and collaboration workflows.

Industry Context

As organizations accelerate digital transformation initiatives, design systems and product velocity have become central to competitiveness. However, despite advances in collaborative design tools, the handoff between design and engineering remains largely manual.

Common challenges include:

- Rebuilding UI components from scratch despite finalized design systems
- Inconsistent interpretation of spacing, responsiveness, and tokens
- Delayed backend alignment with UI data models
- Testing introduced late in development cycles
- Limited traceability between design artifacts and production code

These inefficiencies increase time-to-market, inflate engineering costs, and create quality risks.

Studio Synthesis introduces an automated design-to-code layer that systematically converts structured design artifacts into enterprise-grade application code, embedding reliability and traceability from the outset.

Platform Overview

Studio Synthesis operates through a structured generation pipeline:

1. **Design Ingestion** – The platform connects to Figma files and extracts structured metadata, including layout hierarchy, component structure, design tokens, and interaction logic.
2. **AI Code Generation** – The system translates extracted design data into modular application code, including frontend components, backend APIs, routing logic, and database schemas.

3. **Automated Testing Layer** – Unit and integration tests are generated concurrently with application logic, following a test-driven development (TDD) model.
4. **Repository & DevOps Integration** – Generated artifacts are committed directly into GitHub repositories with structured pull requests and CI compatibility.
5. **Operational Monitoring** – A centralized dashboard tracks job status, completion metrics, and execution logs.

This structured pipeline ensures that design fidelity, engineering standards, and deployment readiness are preserved throughout the generation process.

1.1. Core Capabilities

1.1.1. Full-Stack Application Generation

Studio Synthesis generates production-ready applications rather than static frontend prototypes. Output includes:

- React-based frontend components
- State management and routing configuration
- API endpoints aligned with UI data flows
- Database schemas and validation logic
- Modular, maintainable project structure

The generated code adheres to modern software architecture practices, ensuring scalability and extensibility.

1.1.2. Test-Driven Development Integration

Quality assurance is embedded into the generation process. For each module created, the platform produces:

- Unit test cases
- API validation tests
- Integration test scaffolding

This ensures that applications are delivered with baseline reliability and CI/CD readiness, reducing post-generation QA effort and regression risk.

1.1.3. High-Fidelity Design Implementation

Studio Synthesis preserves pixel-level accuracy by mapping Figma properties directly to implementation-level styling.

This includes:

- Layout constraints and grid systems
- Typography scales and design tokens
- Responsive breakpoints
- Component hierarchy and nesting
- Theming and reusable design patterns

By eliminating interpretation variability, the platform reduces design drift and ensures consistency across releases.

GitHub-Native Workflow

Rather than operating as a closed environment, Studio Synthesis integrates directly into enterprise Git workflows.

Key features include:

- Automatic pull request creation
- Structured commit history
- Branch generation
- CI pipeline compatibility
- Code review enablement

Engineering teams maintain full ownership and visibility of generated artifacts within their existing repositories.

Enterprise Integration Ecosystem

Studio Synthesis is designed to complement established collaboration and project management tools.

Slack Integration

Real-time notifications inform teams of:

- Generation job initiation
- Completion status
- Pull request availability
- Error or failure alerts

This ensures transparent communication within engineering channels.

Jira Integration

The platform links generation activities to Agile workflows by:

- Associating pull requests with Jira tickets
- Updating issue status upon job completion
- Preserving traceability between design artifacts and sprint deliverables

This supports compliance, auditing, and sprint governance requirements.

Microsoft Teams Integration

For organizations operating within Microsoft ecosystems, Teams integration provides structured notifications, enabling cross-functional visibility across engineering, product, and operations teams.

Security and Governance

Studio Synthesis is built with enterprise security principles:

- Role-Based Access Control (RBAC)
- Secure API authentication
- Encrypted communication channels
- Audit logs for generation activity
- Repository-level permission controls

All generated code resides within the organization's Git infrastructure, ensuring data ownership and intellectual property protection.

Operational Transparency

The platform includes a centralized dashboard that provides real-time visibility into generation workflows.

Metrics include:

- Total jobs
- Completed jobs
- Failed jobs
- Pending jobs

Execution logs and status tracking allow teams to identify issues quickly and maintain operational accountability.

Business Impact

Organizations implementing Studio Synthesis report measurable improvements in development efficiency and quality assurance:

- Significant reduction in manual frontend implementation
- Accelerated MVP and feature release cycles
- Improved sprint velocity
- Reduced QA rework
- Stronger alignment between design and engineering

By automating repetitive development tasks while preserving engineering oversight, the platform enables teams to focus on business logic, system architecture, and innovation.

Conclusion

Studio Synthesis represents a strategic advancement in enterprise software development workflows. By automating the transformation of design assets into full-stack, test-enabled, and repository-integrated applications, the platform reduces friction between design and engineering while maintaining governance and quality standards.

It is not positioned as a replacement for engineering teams but as an augmentation layer that increases productivity, reduces implementation variance, and accelerates time-to-market.

For enterprises seeking to modernize delivery pipelines, scale design systems, and improve operational efficiency, Studio Synthesis provides a secure, scalable, and production-ready solution.

2. Testkraft

Autonomous AI-Powered End-to-End Testing Platform

Overview

Testkraft is an enterprise-grade AI-driven testing platform that automates end-to-end software validation across UI, API, Data, and Unit layers. It eliminates manual test scripting, reduces maintenance overhead, and ensures continuous quality across modern application architectures.

Testkraft transforms traditional QA from reactive defect detection into proactive, intelligent quality assurance.

Testing Coverage Across All Layers

Testkraft validates applications across the complete software stack:

UI Testing

- Automated functional UI test generation
- Cross-browser validation
- Dynamic element detection

- Self-healing selectors
- End-to-end user journey coverage

Ensures front-end stability and seamless user experience.

API Testing

- Automated API contract validation
- Schema and payload verification
- Response time monitoring
- Edge case scenario testing
- Microservices interaction validation

Ensures backend services perform reliably.

Data Testing

- Database validation checks
- Data integrity verification
- ETL testing
- Transaction-level validation
- Data consistency across systems

Ensures accuracy and reliability of enterprise data.

Unit Testing

- AI-generated unit test cases
- Code coverage enhancement
- Edge case detection
- Mock dependency simulation
- Automated regression validation

Strengthens code-level quality before integration.

How Testkraft Works

Testkraft operates through an intelligent AI-driven pipeline:

1. Application & Code Analysis

The AI analyzes:

- UI structure
- API contracts

- Database schemas
- Source code components

It understands application behavior before generating tests.

2. Autonomous Test Generation

AI automatically creates:

- UI test flows
- API validation scripts
- Data integrity tests
- Unit test cases

No manual scripting required.

3. Intelligent Test Execution

Tests are:

- Integrated into CI/CD pipelines
- Executed on every build
- Run across multiple environments
- Continuously monitored

4. Self-Healing & Maintenance-Free Testing

When changes occur:

- UI locators auto-update
- API changes are detected
- Schema updates are validated
- Test scripts adapt automatically

Eliminates traditional test maintenance overhead.

5. AI Validation & Insights

Testkraft not only tests — it analyzes:

- Root cause identification
- Defect clustering
- Failure pattern detection
- Coverage gap analysis
- Risk-based prioritization

Provides actionable insights to engineering teams.

Business Impact

- 60–80% reduction in manual QA effort
- Faster release cycles
- Improved defect detection accuracy
- Reduced regression failures
- Lower long-term maintenance costs
- Increased deployment confidence

Enterprise-Ready Architecture

- CI/CD native integration
- Secure SSO & role-based access
- Cloud-scalable execution
- API-first design
- Microservices architecture
- Enterprise compliance support

Competitive Advantage

- Covers UI + API + Data + Unit testing in one platform
- Eliminates test suite decay
- Reduces QA bottlenecks
- Accelerates DevOps pipelines
- Enables continuous quality engineering

Conclusion

Testkraft delivers intelligent, autonomous testing across the full software lifecycle. By combining AI-driven generation, self-healing automation, and full-stack validation, it enables organizations to ship faster, safer, and with greater confidence.

3. GENESIS

Intelligent Document & Underwriting Assistant

Transforming Manual Underwriting into Intelligent, Explainable Decision-Making

Underwriting in banking and insurance is document-intensive, manual, and increasingly complex. Organizations must process large volumes of financial statements, risk reports, medical records, compliance documentation, and customer communications—often scattered across multiple systems.

Genesys is an AI-powered Intelligent Document & Underwriting Assistant that converts fragmented, multi-format data into structured, explainable, and audit-ready underwriting dossiers. It enables institutions to:

- Reduce underwriting cycle time by up to **80%**
- Improve decision consistency and accuracy
- Ensure **100% traceable and auditable outputs**
- Increase underwriter productivity without increasing headcount

The Business Challenge

Today's underwriting teams face:

1. Fragmented Data Ecosystems

- PDFs, scanned documents, Excel sheets, emails
- Data spread across Jira, SharePoint, S3, internal systems

2. Manual Extraction & Reconciliation

- Analysts manually review hundreds of pages
- Time spent summarizing instead of decision-making

3. Slow Turnaround Time

- Cases take days instead of hours
- Customer experience suffers

4. Rising Compliance Pressure

- Regulatory scrutiny increasing
- Difficulty explaining how decisions were derived

The Genesys Solution

Genesys acts as an **AI Co-Pilot for Underwriters**.

It automatically:

- Ingests documents from multiple sources
- Extracts key financial, medical, and risk indicators
- Connects related data across documents
- Builds contextual relationships
- Generates a pre-formatted underwriting dossier
- Maintains full traceability to original source documents

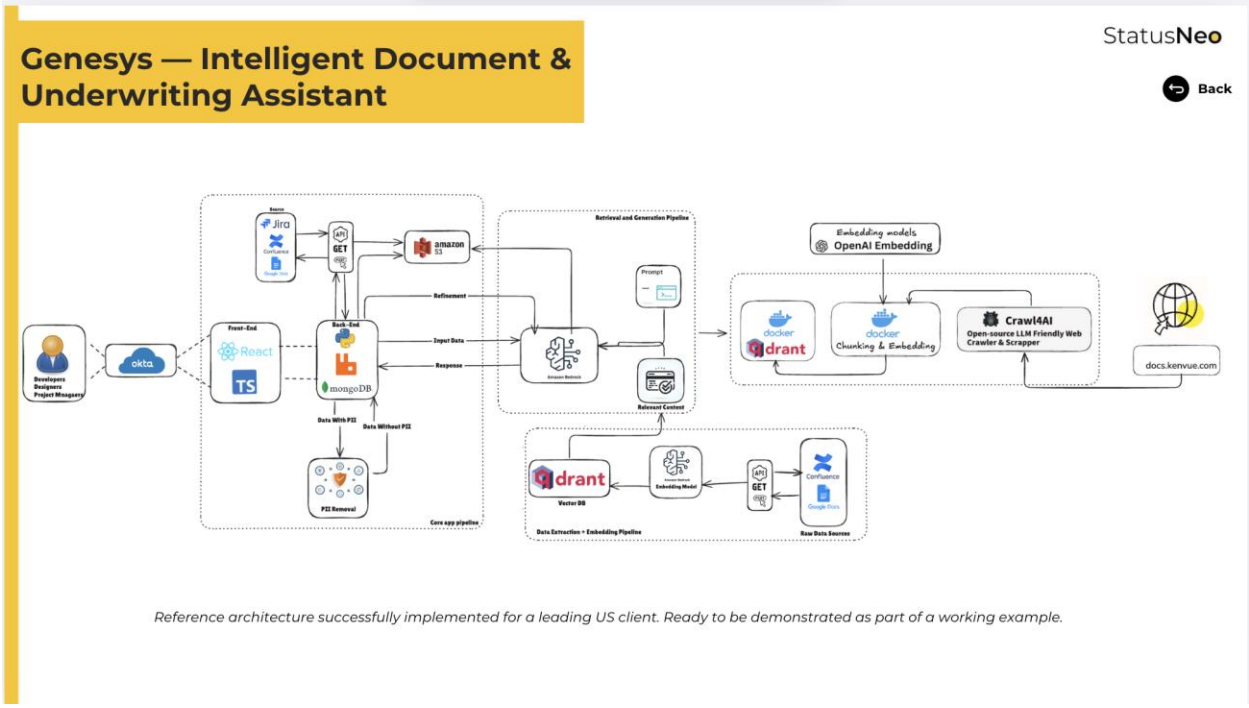
Underwriters receive a **decision-ready report**, not raw documents.

Measurable Business Impact

Metric	Impact
Underwriting Cycle Time	↓ Up to 80%
Decision Accuracy	Significant improvement
Audit Readiness	100% traceable insights
Operational Efficiency	Higher case throughput
Cost Optimization	Reduced manual effort

Solution Overview & Enterprise Readiness

How Genesys Works (Simplified Flow)



Step 1 – Multi-Source Document Ingestion

Objective

Collect all underwriting-related documents from different enterprise systems.

Input Sources

- Document repositories (S3, SharePoint, Drive)
- Case management systems
- Jira / Confluence
- CRM systems
- Email systems
- APIs from internal applications

Technical Process

1. API connectors fetch documents in batch or real-time.
2. Authentication handled via SSO (e.g., Okta).
3. Files stored securely in object storage (segregated by case ID).
4. Metadata captured:
    a. Document type
    b. Timestamp

    c. Source system
    d. Case reference

Output

Raw documents + structured metadata ready for processing.

Step 2 – Document Preprocessing & OCR

Convert all documents into machine-readable format.

Scenarios Handled

- Scanned PDFs
- Handwritten forms
- Image-based documents
- Structured PDFs
- Excel files

Technical Components

- OCR Engines (Tesseract / Textract / Azure Form Recognizer)
- Table extraction modules
- Layout-aware parsing

Processing Steps

1. Document classification (detect document type).
2. If scanned → OCR pipeline.
3. Extract:
    a. Key-value pairs
    b. Tables
    c. Text blocks
4. Normalize extracted text.
5. Convert into structured JSON.

Output

Clean, structured text + layout-aware JSON.

Step 3 – Data Standardization & PII Handling

Ensure security, compliance, and consistent formatting.

Technical Flow

1. PII detection module identifies:
    a. SSN
    b. Account numbers
    c. Personal identifiers
2. Data segregation:
    a. With PII → secure encrypted store
    b. Without PII → analytics pipeline
3. Data normalization:
    a. Standardize dates
    b. Currency normalization
    c. Financial metric formatting

Output

Compliance-ready structured dataset.

Step 4 – Entity & Relationship Extraction (AI Layer)

Objective

Extract meaningful underwriting intelligence.

AI Tasks Performed

- Named Entity Recognition (NER)
- Financial metric extraction
- Risk keyword detection
- Medical terminology detection
- Regulatory markers

Example Extracted Entities

- Applicant name
- Income
- Debt ratio
- Claim history
- Risk score
- Policy number

Technical Flow

1. LLM or NLP model processes document chunks.
2. Extracts entities and context.
3. Assigns confidence score.
4. Stores structured entity graph.

Output

Structured underwriting intelligence.

Step 5 – Knowledge Graph Construction

Objective

Connect all extracted entities across documents.

Why Important?

Underwriting decisions depend on relationships:

- Applicant ↔ Policies
- Applicant ↔ Claims
- Income ↔ Debt obligations
- Risk indicators ↔ Prior decisions

Technical Process

1. Nodes created:
    a. Person
    b. Policy
    c. Claim
    d. Financial metric
2. Relationships defined:
    a. HAS_POLICY
    b. HAS_CLAIM
    c. OWES
    d. ASSOCIATED_WITH
3. Source lineage stored on each node/edge.

Key Feature

Every piece of synthesized insight can be traced back to:

- Document ID

- Page number
- Source system

Output

Graph database representing full case context.

Step 6 – Embedding & Semantic Indexing

Objective

Enable contextual retrieval for LLM synthesis.

Technical Flow

1. Documents are chunked into sections.
2. Each chunk converted into vector embedding.
3. Stored in Vector DB (e.g., Qdrant).
4. Indexed for semantic similarity search.

Why Required?

When generating a report:

- Only relevant content is retrieved.
- Prevents hallucination.
- Improves contextual accuracy.

Output

Searchable semantic memory layer.

Step 7 – Retrieval-Augmented Generation (RAG)

Objective

Generate accurate, explainable underwriting summaries.

Flow

1. Underwriter triggers report generation.
2. System retrieves:
    a. Relevant document chunks (vector search)
    b. Related entities (graph query)
3. Context assembled into structured prompt.
4. LLM generates:

a. Executive summary
b. Risk analysis
c. Financial overview
d. Compliance findings
e. Recommendation

Safety Controls

- Grounded only on retrieved data.
- Citations attached to each insight.
- Confidence scoring.

Output

Draft underwriting dossier.

Step 8 – Lineage Memory & Traceability

Objective

Ensure audit compliance.

Technical Mechanism

Each output paragraph includes:

- Source document reference
- Extracted entity reference
- Timestamp
- Processing layer

Auditors can:

- Click insight
- See original page
- View extraction log

Output

Fully explainable AI output.

Step 9 – Report Formatting & Delivery

Objective

Provide underwriter-ready output.

Supported Formats

- PDF
- Word
- API JSON
- Dashboard view

Structure Includes

- Executive Summary
- Financial Evaluation
- Risk Assessment
- Compliance Check
- Final Recommendation
- Source References

Step 10 – Feedback & Continuous Learning

Objective

Improve system over time.

Process

1. Underwriter edits captured.
2. Differences logged.
3. Feedback fed into:
    a. Prompt refinement
    b. Fine-tuning dataset
    c. Knowledge graph adjustments

Result

Improved accuracy over time.

Security & Enterprise Controls

- SSO (Okta / Azure AD)
- Role-based access control
- Encryption at rest & in transit
- Audit logging
- Multi-cloud deployment

- Containerized microservices

Vertical AI Solutions

1. Status Neo Voice Agent

AI-Powered Multilingual Conversational Telephony Platform

The **StatusNeo Voice Agent** is an enterprise-grade AI telephony solution that automates customer conversations using real-time speech processing and conversational AI.

It enables businesses to conduct intelligent inbound and outbound calls with natural, human-like voice interactions across multiple languages.

How It Works

The Voice Agent operates through a real-time conversational pipeline:

1. **Call Initiation** – Customer connects via Twilio.
2. **Audio Streaming** – Real-time bidirectional audio via secure WebSocket.
3. **Speech-to-Text (STT)** – Customer speech converted to text.
4. **AI Processing** – GPT-based engine understands intent and context.
5. **Response Generation** – AI creates a goal-oriented response.
6. **Text-to-Speech (TTS)** – Response converted into natural voice.
7. **Playback** – Customer hears the AI agent instantly.

Low latency ensures smooth, natural conversation.

Core Capabilities

Multilingual Support

Supports major Indian and international languages with regional accent handling and real-time processing.

Intelligent Conversation Management

- Maintains conversation context
- Handles multi-intent interactions
- Manages natural turn-taking
- Adapts tone based on sentiment

Persona-Based Interactions

Customizable personas aligned with business use cases:

- Banking collections
- Healthcare follow-ups
- Customer service interactions

Each persona includes defined tone, goal, and conversational style.

Business Functions

The Voice Agent can automate:

- Payment reminders and collections
- Appointment scheduling and confirmations
- Customer support interactions
- Lead qualification
- Follow-up calls

Industry Applications

Banking & Finance

- EMI reminders
- Loan follow-ups
- Fraud alerts

Healthcare

- Appointment reminders
- Medication adherence calls

Insurance

- Policy renewals
- Claim follow-ups

E-commerce

- Delivery updates
- Payment reminders

Enterprise Architecture

- Twilio telephony integration
- AI conversation engine (GPT-based)

- Real-time speech processing
- Scalable cloud deployment
- Secure access controls

Business Impact

- Reduced call center operational costs
- Increased engagement and recovery rates
- 24/7 automated customer interaction
- Scalable multilingual communication
- Consistent, goal-driven conversations

Conclusion

The StatusNeo Voice Agent transforms traditional IVR systems into intelligent, conversational AI agents capable of handling real-time, multilingual, business-critical customer interactions at scale.

2. AML & Transaction Monitoring Agent

Graph-based AI detection of complex money laundering patterns.

Impact: Enhanced fraud detection, reduced false positives, improved compliance efficiency **Maturity:** Production

The **AML Transaction Monitoring Agent** is an enterprise-grade AI solution designed to detect, analyze, and flag suspicious financial transactions in real time. It combines advanced pattern detection algorithms, graph-based analysis, and intelligent risk scoring to help financial institutions strengthen compliance and reduce financial crime exposure.

The platform enables compliance teams to move from reactive investigation to proactive risk detection.

Data Connectivity & Integration

The system supports flexible data integration to accommodate enterprise environments:

- **Upload CSV / Excel files** for batch analysis
- **Connect to PostgreSQL databases** for direct transactional monitoring
- **Integrate with external APIs** for real-time streaming data

This ensures seamless deployment across legacy systems, core banking platforms, and fintech infrastructures.

How the AI Agent Detects Money Laundering

The detection engine operates through a structured 5-step pipeline:

1. Data Input

Transaction data is ingested from:

- CSV files
- Databases
- APIs

Data is validated, normalized, and prepared for analysis.

2. Pattern Detection Engine

The AI engine runs 10+ detection algorithms designed to identify complex laundering behaviors.

It uses:

- Rule-based checks
- Behavioral analysis
- Graph relationship modeling
- Machine learning-based anomaly detection

3. Risk Analysis

Each transaction or entity is assigned:

- A severity level
- A calculated risk score
- Contextual relationship insights

This enables prioritization based on potential compliance exposure.

4. Alert Generation

Suspicious patterns trigger:

- Real-time alerts
- Structured investigation reports
- Audit-ready documentation

Alerts are designed to minimize false positives while ensuring regulatory coverage.

5. AI Assistant for Investigation

The built-in AI assistant supports compliance officers by:

- Explaining why a transaction was flagged
- Summarizing risk pathways
- Highlighting related accounts and transaction chains
- Accelerating case resolution

Patterns Detected

The system detects advanced money laundering schemes, including:

- **Round-Tripping** – Funds transferred and returned to conceal origin
- **Funnel Accounts** – Multiple deposits consolidated into a single account
- **Multi-Tier Layering** – Complex multi-step transaction masking
- **Synthetic Identity Fraud** – Fabricated identities for illicit activity
- **U-Turn Loans** – Circular loan structures for concealment
- **Money Circles** – Structured circular transfer loops
- **Scatter-Gather Schemes** – Funds split and recombined to avoid detection
- **Crypto Mixing** – Blockchain-based laundering mechanisms

These go beyond traditional rule-based AML systems by identifying network-level behaviors.

Business Impact

The AML Transaction Monitoring Agent delivers measurable value:

- Faster detection of suspicious activity
- Reduced false positives
- Lower investigation time per case
- Improved regulatory compliance
- Enhanced risk transparency
- Real-time monitoring capabilities

Enterprise-Ready Architecture

- Secure, role-based access control
- Scalable cloud-native deployment
- API-first integration model
- Audit trail for every flagged transaction
- Designed for banking, fintech, and financial institutions

Conclusion

The AML Transaction Monitoring Agent transforms traditional compliance monitoring into an intelligent, AI-driven risk detection system. By combining real-time pattern analysis, graph-based intelligence, and explainable alerts, it empowers compliance teams to detect complex laundering schemes with greater speed, accuracy, and confidence.